

ORIGINAL ARTICLE

Income differences in stroke mortality: A 12-year follow-up study of the Swedish working population

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Abstract

Aims: This study explored the association between income and stroke mortality in the total working population in Sweden and examined whether the associations differ by gender or for stroke subtypes intracerebral haemorrhage (ICH) or brain infarction (BI). **Methods:** This was a register-based study among nearly 3 million working women and men (30–64 years in 1990) with a 12-year follow up (1991–2002) for mortality from stroke (4886 deaths). Income was measured as annual registered income from work in 1990. Gender-specific Cox regressions were applied with adjustments for sociodemographic covariates. **Results:** The age-adjusted hazard ratio (95% confidence interval) of lowest versus highest income quartile was 1.80 (1.48–2.19) for any stroke, 1.68 (1.29–2.17) for ICH and 2.23 (1.53–3.22) for BI in women, and the corresponding figures for men were 2.12 (1.92–2.34), 2.02 (1.77–2.31), and 2.09 (1.77–2.46). Adjustment for covariates attenuated these associations to 1.69 (1.33–2.15) for any stroke and 1.56 (1.14–2.14) for ICH in women and to 1.98 (1.74–2.24) for any stroke and 1.77 (1.44–2.19) for BI in men. In contrast, adjustment for covariates amplified the estimates to 2.36 (1.52–3.66) for BI in women and to 2.05 (1.73–2.44) for ICH in men. **Conclusions:** Risk of stroke mortality was highest in the lowest income group, with a gradient for the intermediate groups, in both women and men. The risk of mortality from BI was highest in women with the lowest income and the risk of ICH was highest in men with the lowest income.

Key Words: Epidemiology, income, mortality, socioeconomic position, stroke

Introduction

Stroke is among the leading causes of death and long-term disability worldwide [1]. In Sweden, roughly 30,000 people suffer from stroke each year, and about a fifth of them are of working age (<65 years) [1]. The consequences of stroke with regard to production loss, sickness absence, disability pension, and premature death are considerable [2,3]. Yet, only a few studies have focused on income and other work-related factors in relation to the risk of stroke among working-age populations [4–8]. An income gradient – where those with a lower income have a higher risk of stroke than those with a higher income – has been established for stroke incidence and mortality [4–8], and recurrence and case fatality [8]. However, there is still some uncertainty as to whether an income gradient is present both in

employed women and in men and for the main stroke subtypes intracerebral haemorrhage (IHR) or brain infarction (BI). A recent review study shows that stroke is more common among men, but women suffer from more severe types of stroke and case fatality is higher among women (i.e. the risk of dying after suffering a stroke) [9]. Also, the influence of risk factors on stroke subtypes may differ as, for instance, severe psychological stress was only associated with non-haemorrhagic stroke in men in Sweden [10].

In line with previous findings on income and stroke [4–8], the present study hypothesised that the risk of mortality from stroke is higher for working women and men with low income. However, the association between income and the risk of stroke may differ by gender and also by stroke subtype (ICH or BI) as previous studies are inconclusive [5–8,11,12]. Based on register data on the total working population in

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Sweden, the present study thus aimed to: (1) establish the association between income and mortality from stroke, properly adjusted for demographic, socioeconomic, and work-related factors, and (2) explore whether the association differs by gender or for stroke subtypes.

Materials and methods

Study population

The data came from the Swedish Work and Mortality Database maintained at the Centre for Health Equity Studies, Stockholm University/Karolinska Institute. The database compiles several national registers provided by Statistics Sweden, including the Total Population Register (TPR), National Population and Housing Censuses, and the Longitudinal Database on Education, Income and Employment (LOUISE), all of which were used in the present analyses. Linkages were possible using the 10-digit personal identity number, which was replaced by a serial number to ensure anonymity.

All people in Sweden who were 30–64 years of age in 1990 and had a job (employees and self-employed) at the time of the census were included ($N=2,991,973$). People were excluded from the analyses if they had a prior stroke diagnosis in the Hospital Discharge Register in 1980–1990 ($n=4167$) and if they had emigrated during the follow up as recorded in the TPR ($n=41,898$). Persons with missing values on central variables, as well as those with zero registered annual income from work, were excluded. The final cohort consisted of 2,825,117 people (49.5% women).

The cohort was followed-up for mortality from stroke by record linkage to the Cause of Death Register for the period 1991–2002. Each participant was considered at risk from the beginning of the follow up (1 January 1991) to death from stroke, death from another cause, or end of follow up (31 December 2002). There were 16,523,480 female and 16,714,797 male person-years in the follow-up period.

Income from work

Income was measured in 1990 as individual income from work, which comprises taxable annual income including wages, self-employed income, and taxable income transfers (e.g. sickness insurance). The income measure was classified into four groups by quartile values of the income distribution in the combined data for women and men, and those in the

highest income group were used as the reference category.

Covariates

Age was treated as a continuous variable in the multivariate analyses. Marital status was classified into four categories: married (reference category), never married, divorced, and widowed. As differences in stroke incidence and stroke subtypes have previously been reported between immigrants and natives [13], the present analyses were adjusted for country of birth (foreign born, yes/no). Educational level was classified into four categories: unknown, primary, secondary, or tertiary (reference category). Occupational class was classified into four groups: manual workers (unskilled and skilled workers), routine non-manuals, intermediate and higher non-manuals (reference category), and self-employed. Job control, a combined measure of decision authority (four items) and skill discretion (three items) at work, was aggregated to occupations in the 1990 census using a psychosocial job exposure matrix (JEM) [12]. Briefly, the JEM is based on occupational titles and the information is thus not self-reported. Job control was divided into four groups, and those in the highest job control group were used as the reference category. As fewer women than men work full time, the analyses were adjusted for a dichotomous variable for work time (working full time, yes/no). All these variables were measured at the baseline in 1990.

Stroke endpoints

Mortality from stroke was defined by the following underlying causes of death: intracerebral haemorrhage (ICH: ICD-9, 431; ICD-10, I61), cerebral (brain) infarction (BI: ICD-9-434; ICD-10, I63), and undetermined pathological type (UND: ICD-9, 436; ICD-10, I64). First, any type of stroke was studied that included ICH, BI, and UND. Then, ICH and BI were analysed separately.

Statistical analyses

Gender-specific analyses were performed, as the income distribution differed markedly by gender (Tables I and II). Age-standardised mortality rates of stroke subtypes by income groups were calculated per 100,000 person-years at risk (Table III) [14]. Direct standardisation was applied using 5-year age groups and the age-specific person-years in the data as the standard population. This method controlled for differences in the age distribution between women and men, and between the income groups.

Table I. Working women in Sweden, 1990: baseline characteristics by income group.

Characteristic	Income groups			
	Lowest	2nd	3rd	Highest
Women ($n = 1,397,084$)	37.8	33.4	18.8	10.0
Income (1000s SEK, median $\pm$ SD)	69 $\pm$ 27	120 $\pm$ 12	157 $\pm$ 12	209 $\pm$ 49
Age (years, mean $\pm$ SD)	44.3 $\pm$ 10.1	44.7 $\pm$ 8.7	45.3 $\pm$ 8.2	46.4 $\pm$ 7.7
Marital status				
Married	72.0	66.6	60.2	62.6
Never married	14.5	16.9	20.5	18.6
Divorced	9.5	13.6	17.0	16.6
Widowed	4.0	2.9	2.3	2.2
Foreign born (yes)	11.1	10.0	9.6	8.0
Education				
Unknown	0.2	0.1	0.09	0.08
Primary	38.0	30.6	20.5	9.5
Secondary	45.9	49.5	40.0	21.4
Tertiary	15.9	19.8	39.4	69.0
Occupational class				
Manual	58.6	50.2	22.8	6.1
Lower non-manual	20.7	26.3	29.6	14.3
Higher non-manual	16.3	22.5	46.8	78.7
Self-employed	4.4	1.0	0.8	0.9
Job control				
Lowest	42.9	33.4	21.2	11.8
Low intermediate	28.9	34.2	34.0	38.3
High intermediate	22.1	25.5	30.9	25.4
Highest	6.1	6.9	13.9	24.5
Working full time (yes)	23.4	60.0	86.4	89.7

Values are % unless otherwise stated.

Table II. Working men in Sweden, 1990: baseline characteristics by income group.

Characteristic	Income groups			
	Lowest	2nd	3rd	Highest
Men ($n = 1,428,033$)	12.6	16.8	31.0	39.6
Income (1000s SEK, median $\pm$ SD)	58 $\pm$ 31	128 $\pm$ 12	161 $\pm$ 12	226 $\pm$ 93
Age (years, mean $\pm$ SD)	46.6 $\pm$ 10.5	44.2 $\pm$ 9.9	44.0 $\pm$ 9.1	45.2 $\pm$ 8.3
Marital status				
Married	58.6	56.4	63.9	75.0
Never married	28.4	32.4	26.2	16.0
Divorced	11.8	10.3	9.1	8.4
Widowed	1.2	0.9	0.8	0.6
Foreign born (yes)	16.4	11.7	8.9	6.7
Education				
Unknown	0.5	0.3	0.1	0.1
Primary	44.2	45.9	41.2	18.2
Secondary	42.3	43.1	45.7	39.0
Tertiary	13.0	10.7	13.0	42.7
Occupational class				
Manual	55.0	69.9	63.2	20.3
Lower non-manual	8.8	11.9	11.8	10.9
Higher non-manual	14.2	14.5	23.0	66.5
Self-employed	22.0	3.7	2.0	2.3
Job control				
Lowest	25.4	26.9	23.2	8.2
Low intermediate	23.5	27.3	22.2	9.1
High intermediate	30.1	26.6	26.7	18.4
Highest	21.0	19.2	27.9	64.3
Working full time (yes)	61.8	89.7	96.3	96.7

Values are % unless otherwise stated.

Table III. Working women and men in Sweden 1990 ($N = 2,825,117$). Number of deaths from stroke 1991–2002 and age-standardised mortality rates per 100,000, by income group.

Income	Women ($n = 1,397,084$)						Men ($n = 1,428,033$)					
	Any stroke ^a			ICH ^b			Any stroke ^a			ICH ^b		
	Events (n)	Mortality rate	Events (n)	Events (n)	Mortality rate	Events (n)	Events (n)	Mortality rate	Events (n)	Events (n)	Mortality rate	Events (n)
Highest group	113	6.94	66	31	1.96	851	488	13.72	296	488	7.70	296
2nd group	237	8.27	139	75	2.67	862	464	18.75	317	464	9.82	317
3rd group	485	9.33	292	154	3.00	731	351	26.55	299	351	12.91	299
Lowest group	796	12.30	413	287	4.26	811	406	30.84	298	406	16.73	298
Total no. of events	1631	9.97	910	547	3.35	3255	1709	19.80	1210	1709	10.37	1210

^aAny stroke: ICD-9, 431, 434, 436; ICD-10, I61, I63, I64.^bIntracerebral haemorrhage: ICD-9, 431; ICD-10, I61.^cBrain infarction: ICD-9, 434; ICD-10, I63.

To establish the associations between income and stroke mortality, hazard ratios (HR) with 95% confidence intervals (95% CI) were calculated using Cox proportional hazards models with length of follow up as the time variable [14]. The log-rank test was used to formally test if the differences in survival during follow up, by income group stratified on gender, were statistically significant. The association between income and stroke mortality was adjusted for age in the base model, and for demographic (marital status and foreign born), socioeconomic (educational level and occupational class), and work-related factors (job control and working full time) in the subsequent model. To test for linear trends in the associations between income and risk of stroke, the median values for income quartiles were treated hierarchically in the Cox proportional hazards models.

Results

Baseline characteristics

Almost 38% of working women in Sweden were in the lowest income group in 1990 (Table I). These women were slightly younger and a larger proportion of them were married and foreign born as compared with women in the higher income groups. The proportion of tertiary education was lowest in the lowest income group and more of these women were manual workers and self-employed. They also had the highest share of low job control and only 23.4% of them worked full time. However, the majority of women in Sweden who worked part time spent 30 hours rather than 15 hours per week in paid work [15].

Of the working men in Sweden, about 13% were in the lowest income group (Table II). These men were older than the men in the higher income groups and a smaller proportion of them were married. Also, the proportions of never married and divorced men were highest among men in the lowest income group. The share of tertiary education was between 11 and 13% in the three lowest income groups for men. For women, the share of tertiary education increased markedly by income and 69% of women in the highest income group had a tertiary education compared to 43% of men. For men, the share of low job control was of similar magnitude in the three lowest income groups. About 62% of men in the lowest income group worked full time compared to 97% of men in the highest income group.

Table IV. Working women and men in Sweden 1990 ($N = 2,825,117$). Hazard ratios (HR) and 95% confidence intervals (95% CI) for mortality from stroke or subtypes by income groups, adjusted for confounders at baseline: follow up 1991–2002.

	Women ($n = 1,397,084$)				Men ($n = 1,428,033$)			
	Any stroke ^a	ICH ^b	BI ^c		Any stroke ^a	ICH ^b	BI ^c	
Model 1								
Highest income group	1	1	1		1	1	1	
2nd income group	1.21 (0.9–71.51)	1.21 (0.90–1.62)	1.39 (0.92–2.12)		1.35 (1.23–1.49)	1.28 (1.13–1.45)	1.42 (1.21–1.67)	
3rd income group	1.39 (1.13–1.70)	1.44 (1.1–01.88)	1.58 (1.07–2.32)		1.91 (1.73–2.11)	1.66 (1.45–1.91)	2.17 (1.84–2.55)	
Lowest income group	1.80 (1.4–82.19)	1.68 (1.29–2.17)	2.23 (1.53–3.23)		2.12 (1.92–2.34)	2.02 (1.77–2.31)	2.09 (1.77–2.46)	
Model 2								
Highest income group	1	1	1		1	1	1	
2nd income group	1.07 (0.84–1.35)	1.07 (0.79–1.45)	1.27 (0.82–1.95)		1.20 (1.08–1.33)	1.21 (1.05–1.39)	1.14 (0.96–1.36)	
3rd income group	1.22 (0.97–1.54)	1.26 (0.94–1.70)	1.52 (1.00–2.32)		1.63 (1.46–1.83)	1.54 (1.32–1.80)	1.66 (1.38–2.00)	
Lowest income group	1.69 (1.33–2.15)	1.56 (1.14–2.14)	2.36 (1.52–3.66)		1.98 (1.74–2.24)	2.05 (1.73–2.44)	1.77 (1.44–2.19)	
Test for trend	$p = 0.000$	$p = 0.001$	$p = 0.000$		$p = 0.000$	$p = 0.000$	$p = 0.000$	
Number of events	1631	910	547		3255	1709	1210	

Model 1: adjusted for age.

Model 2: adjusted for age, marital status, foreign born, educational level, occupational class, job control, and working full time.

^aAny stroke: ICD-9, 431, 434, 436; ICD-10, I61, I63, I64.^bIntracerebral haemorrhage: ICD-9, 431; ICD-10, I61.^cBrain infarction: ICD-9, 434; ICD-10, I63.

Stroke mortality

During the follow up, 1631 deaths from stroke occurred in women, and 3255 in men (Table III). Of these deaths, 910 were ICH, 547 BI, and 174 UND in women, and 1709 ICH, 1210 BI, and 336 UND in men. Age-standardised mortality rates of stroke and stroke subtypes were higher in the lowest income quartile than in the highest quartile both in women and in men.

Association between income and stroke mortality

Women and men in the lowest income group had a higher risk of mortality from stroke and stroke subtypes than did those in the highest income group, and there was a gradient by income in the two intermediate groups (Table IV, Model 1). Adjustment for potential confounders generally decreased the estimates in both women and men, except for BI in women and ICH in men in the lowest income group (Table IV, Model 2). The risk of mortality from BI (HR 2.36; 95% CI 1.52–3.66) was higher than the risk of mortality from ICH (HR 1.56; 95% CI 1.14–2.14) in women in the lowest income group, while the opposite was seen in men; the risk of ICH (HR 2.05; 95% CI 1.73–2.44) was higher than the risk of BI (HR 1.77; 95% CI 1.44–2.19). The *p*-values for the trend between income and mortality from stroke were significant for all stroke outcomes in both women and men.

Discussion

To our best knowledge, this is the first large-scale register-based follow-up study of income and risk of stroke mortality in a working population in Scandinavia. The risk of stroke mortality was higher in women and men with the lowest income than in those with the highest income, with a gradient for the two intermediate income groups. When analysing income in relation to stroke subtypes, gender differences appeared. The risk of BI was highest in women with the lowest income, and the risk of ICH was highest in men with the lowest income compared to women and men with the highest income.

Theoretical and methodological aspects of studies on income and mortality

Previous studies of income and stroke tend to use household disposable income as the main income measure, thus assuming that income translates to consumption possibilities and material circumstances, which consequently have an effect on

stroke morbidity or mortality. Individual income from work, on the other hand, is part of the socially organised exchange process in which people's work efforts are rewarded in terms of money and esteem [16]. Instead of being merely an indicator of material circumstances, income from work approximates individuals' relative economic status on the labour market [15,17]. Therefore, in addition to the direct effect of material circumstances, income from work may be linked to the risk of stroke also through the physiological effects of its emotional and social meaning [18]. Income from work is correlated with both positive and negative working conditions, and the accumulation of adverse work exposures associated with low income may increase the risk of stroke among working populations [11,12].

A general pitfall in studies of income and mortality is that interpretation of the association may be hampered by influence from other socioeconomic circumstances causally preceding income, such as education and occupational class [19]. Education, occupational class, and income represent different stages in the process of stratification in a society, and they are also likely to affect health through different causal mechanisms [20]. Previous studies conclude that income has an independent effect on health even when education and social class are adjusted for [21]. Another issue of concern is reverse causation, as income may be determined by pre-existing ill health [22]. Especially in working populations, poor health may lead to long-term sick leave or early retirement and thus to a consequent loss of income. The effects of other socioeconomic circumstances may be somewhat reduced when analyses are adjusted for socioeconomic factors that precede income. Reverse causation may be reduced by restricting the analyses to employed populations and by using an appropriate income measure.

The present findings in relation to previous studies

In general, the present study corroborated previous findings showing that the risk of stroke varies across income groups [4–8,23]. There are, however, some important differences regarding the results for gender and stroke subtypes. In the present study, the risk of mortality from BI or ICH was increased both in women and in men with the lowest incomes, but the relative risk was highest for BI in women and for ICH in men. In Finland, income was significantly associated with ischaemic and haemorrhagic stroke in men, but no significant associations were found in women [5,6]. In a population-based study from Sweden, income was significantly associated with ischaemic subtype of stroke, but not haemorrhagic, both in

women and in men [8]. However, these previous studies did not adjust the association between income and stroke for socioeconomic factors that precede income, nor were they restricted to working populations.

Even if women generally experience poorer socioeconomic circumstances than men (except for educational level), they almost always have higher life expectancies, but they also suffer more morbidity. Regarding socioeconomic gradients in health, the associations are usually weaker and flatter in women, and men's commonly found steeper gradients are sometimes suggested to manifest a "general pattern" of health inequalities. However, relative socioeconomic differences in mortality among women and men have been found to be sensitive to the choice of measure for socioeconomic position [24]. Among the working population of the present study, income differences in stroke mortality were of similar magnitude among men and women, and the risk of stroke mortality was even higher in low-income women than in men as regards to one of the stroke subtypes.

Possible underlying mechanisms linking income and stroke

Stroke is a multifactorial disease, and several social factors related to low income along the life course may influence the risk of stroke in adulthood. Reduced fetal growth, adverse childhood circumstances, and adverse influences linked to low income in adulthood are of particular importance [25,26]. Income may be linked to the risk of mortality from stroke partly through a direct consumption effect or a direct status effect [21]. Low income as an indicator of social status is associated with greater psychological distress, and therefore income may be linked to the risk of stroke through the physiological effects of its emotional and social meaning [18]. But it is also likely that low income is linked indirectly to stroke mortality through the accumulation of factors that increase mortality in general [19]. Such an accumulation may be related to adverse occupational careers and work environment exposures.

Strengths and weaknesses of the present study

The present findings are more reliable than those of previous studies on income differences in stroke morbidity and mortality for several reasons. First, register data on the total working population included employed and self-employed women and men with various occupations which make the association between income and risk of stroke mortality valid for all working women and men regardless of

occupation. Second, income was based on an objective register-based measure that was not influenced by self-report bias. Third, information on stroke mortality was assessed from the Cause of Death Register. A validation study from Sweden, which compared routine registers with the population-based MONICA stroke register, concluded that official mortality statistics give a reasonably good estimate of fatal stroke events in the population [27]. Fourth, in order to reduce potential influence from other socioeconomic circumstances, the analyses were adjusted for socioeconomic indicators that precede income. Fifth, reverse causality was reduced by restricting the analyses to the employed population.

One shortcoming of register data is the lack of information on traditional risk factors for stroke. However, traditional risk factors tend to explain only part of the social gradient in the relative risk of cardiovascular diseases [28]. Another weakness of the present study is that income was measured only at the baseline in 1990. Previous studies show that average income is more important for health than is current income; persistent poverty is more harmful than occasional episodes of low income; income falls appear to have greater effect on health than do income increases; and, finally, income level appears to be more important for health than income change [29]. Therefore, to assess the duration of exposure to low income, several measuring points in time are needed. However, one could assume that the income levels of the working population in Sweden do not change dramatically over time as the welfare system provides support when people's labour market participation is challenged. Another source of potential bias is the healthy worker effect, but the influence of this bias would most likely decrease the estimates of the present study as working populations are normally healthier than general populations. Also, the healthy worker effect could be stronger among women than in men so that more men than women maintain their labour market position after falling ill. Previous analyses show that the healthy worker effect is stronger in women than in men for acute myocardial infarction, but not for stroke [30].

Conclusions

People with the lowest income have a considerable higher risk of mortality from stroke than those with higher income. The association between income and risk of mortality from stroke subtypes varies as a function of gender. The relative risk of mortality from brain infarction is highest for the women with the

lowest income. The implication for future studies is that women and men need to be analysed separately, and stroke subtypes need to be taken into account as well. Reducing the excess mortality from stroke in people with low income could markedly decrease the burden of stroke in working populations and thus represent an important public health improvement.

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